BFOR516 - Unsupervised Machine Learning

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Unsupervised Machine Learning

- Unsupervised learning is a type of machine learning where the algorithm learns from data without any labels or predefined answers.
- Unlike supervised learning, where the algorithm is trained using input-output pairs (like images labeled as "cat" or "dog"), unsupervised learning only gets the input data and must find patterns or structures on its own.

Types of Unsupervised Learning

1. **Transformations of Data**
   a. This means changing the way data is represented to make it easier to understand.
   b. One common method is dimensionality reduction, which simplifies data by keeping only the most important features.
   c. **Example:** If you have a dataset with 100 features (columns), dimensionality reduction might reduce it to just 2 features, making it easier to visualize in a graph.
   d. Another example is topic extraction in text analysis, where an algorithm groups words into topics (like "sports," "politics," or "entertainment") without being explicitly told what the topics are.

Types of Unsupervised Learning

1. **Clustering**
   a. Clustering groups similar data points together.
   b. **Example**: A social media app might group your photos based on the faces in them, even though it doesn't know who the people are. It simply finds similar-looking faces and puts them in the same group.

Challenges in Unsupervised Learning

- Since there are no labels, it's hard to measure success. We don't always know if the groups or patterns found by the algorithm are useful.
- Sometimes, the algorithm may create groups that don't make sense for our needs (e.g., grouping pictures by lighting conditions instead of by person).
- Unsupervised learning is often used for exploration—to understand data better before using it in other machine learning tasks.

K-Means Clustering

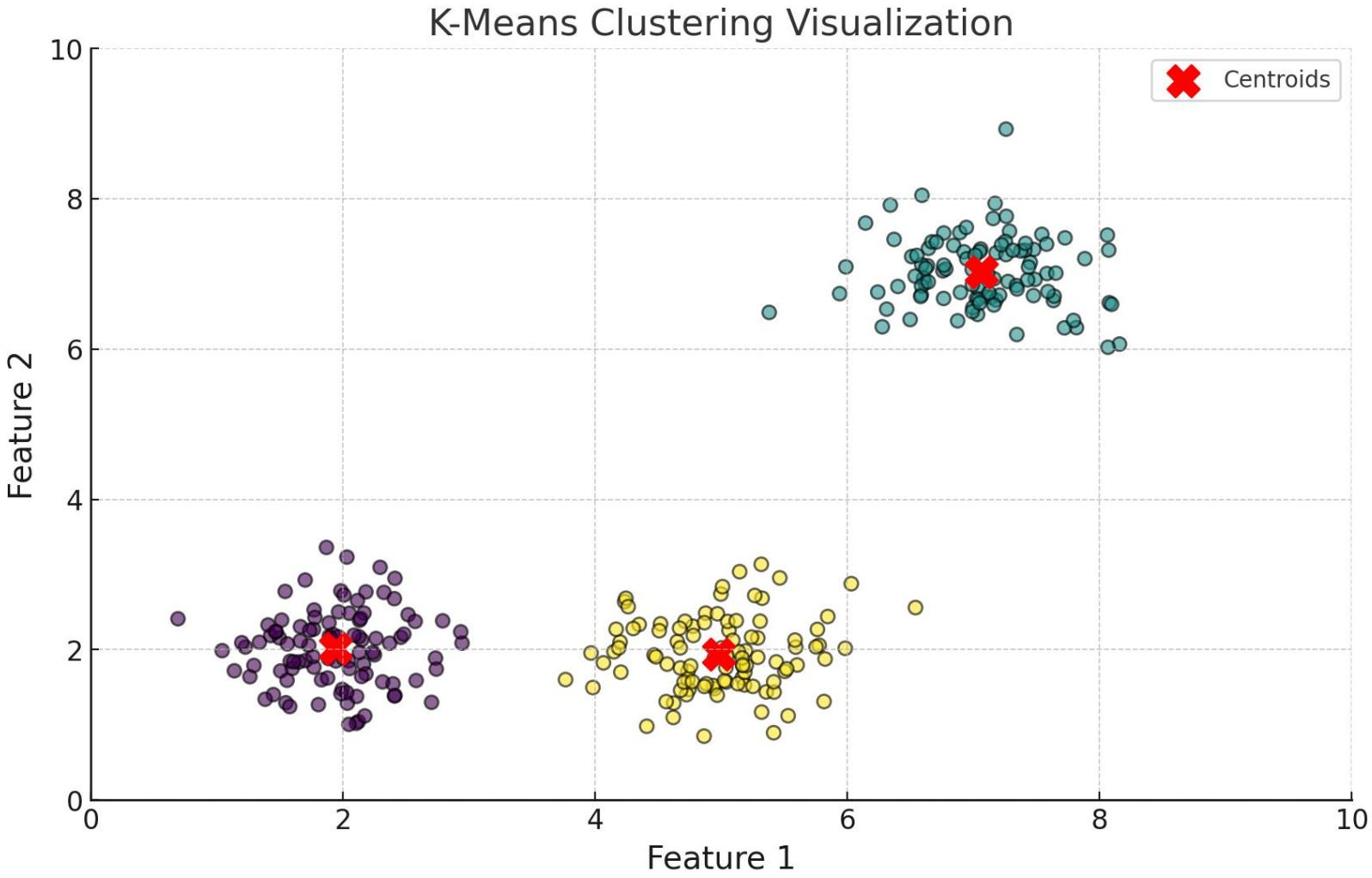
K-Means Clustering Overview

A partitioning method that divides data into k distinct clusters based on distance to the centroid of a cluster.

- Requires the number of clusters (k) to be specified in advance.
- Iteratively assigns data points to clusters and updates cluster centroids.

The K-Means Clustering Process

1. Initialization:
   a. Choose k initial centroids (randomly or using heuristics).
2. Assignment Step:
   a. Assign each data point to the nearest centroid.
3. Update Step:
   a. Calculate the new centroids as the mean of all data points assigned to each cluster.
4. Repeat:
   a. Continue the assignment and update steps until convergence (no change in centroids).

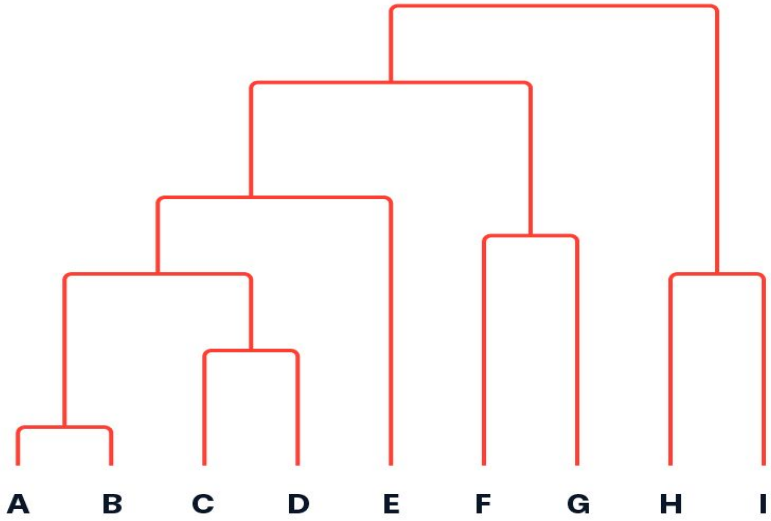

K-Means Clustering Visualization
Centroids
Feature 2
Feature 1
0
2
4
6
8
10
0
2
4
6
8
10

Hierarchical Clustering

Hierarchical Clustering

- Hierarchical clustering is a method of grouping data points into clusters in a step-by-step manner.
- It is like creating a family tree of clusters, where similar data points get grouped together as you move up or down the tree.

Hierarchical Clustering



Types of Hierarchical Clustering

1. Agglomerative (Bottom-Up Approach)
   a. Starts with each data point as its own cluster.
   b. Merges the two most similar clusters at each step.
   c. Stops when all points form a single large cluster.
2. Divisive (Top-Down Approach)
   a. Starts with all data points in one big cluster.
   b. Splits the most different points into separate clusters.
   c. Continues until each point is its own cluster.

How Agglomerative Hierarchical Clustering Works

1. Start: Each data point is treated as an individual cluster.
2. Find the Closest Clusters: Measure the similarity (distance) between all clusters.
3. Merge: Combine the two closest clusters into one.
4. Repeat: Keep merging until only one cluster remains or until a set number of clusters is reached.

How to Measure Distance Between Clusters?

When merging clusters, we need to decide how to measure the distance between them. There are three common ways to do this:

1. Single Linkage Method
   a. Measures the minimum distance between points in two clusters.
   b. Can create long, chain-like clusters.
2. Complete Linkage Method
   a. Measures the maximum distance between points in two clusters.
   b. Creates compact and evenly sized clusters.
3. Average Linkage Method
   a. Measures the average distance between all points in two clusters.
   b. A balance between single and complete linkage.
   c. Centroid Linkage Method: Measures the distance between centroids (the average position of all points in a cluster).

Ward's Method

- Unlike single, complete, and average linkage, Ward's method does not directly use distances.
- Instead, it minimizes the variance within clusters when merging them.
- It often results in balanced, well-separated clusters, making it a popular choice.

Method	Distance Calculation	Effect on Clustering
Single Linkage	Minimum distance between points in two clusters	Tends to form long, chain-like clusters
Complete Linkage	Maximum distance between points in two clusters	Produces compact, evenly sized clusters
Average Linkage	Average distance between all pairs of points in two clusters	Balances between single and complete linkage
Centroid Linkage	Distance between cluster centroids (mean of all points)	Can sometimes cause inversions in dendrograms
Ward's Method	Minimizes variance within clusters rather than distance	Produces compact and well-separated clusters

DBSCAN

DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a clustering algorithm that groups together points that are close to each other in dense regions while identifying outliers (noise).

Why is DBSCAN Useful?

- No need to specify the number of clusters (unlike K-Means).
- Can find clusters of different shapes (not just circular clusters like K-Means).
- Detects outliers (points that don't fit into any cluster).

How does DBSCAN Work

1. Find Dense Areas
   a. DBSCAN looks at each point and checks how many other points are within a certain distance (eps).
   b. If enough points (min_samples) are within this distance, the point is called a core point and forms a cluster.
2. Expand the Cluster
   a. Other nearby points are added to the same cluster if they are within eps distance.
   b. This process continues, growing the cluster outward.
3. Identify Noise
   a. If a point doesn't have enough neighbors within eps, it is marked as noise (not part of any cluster).

Types of Points in DBSCAN

1. **Core Points**: Have at least min_samples points within eps distance.
2. **Boundary Points**: Not core points but close to at least one core point.
3. **Noise Points**: Not close enough to any cluster (assigned label -1).

Key Parameters

1. **eps (ε):** Maximum distance to consider a point as a neighbor.
2. **min_samples:** Minimum number of points needed to form a dense region (a core point).

- If eps is too small → Many points will be labeled as noise.
- If eps is too big → Separate clusters may merge into one large cluster.
- If min_samples is too high → Small clusters may disappear.

DBSCAN Example

Given Dataset:

We have the following points:

A, B, C, D, E, F, G, H, I

Let's assume the following neighborhood relationships exist within distance eps (ε):

Point	Neighbors within ε	Identify Points
A	B	❌Not a core point.
B	A, C, G	✅Core Point
C	B	❌Not a core point.
D	E,F	❌Not a core point.
E	D,F	❌Not a core point.
F	D,E	❌Not a core point.
G	H,I	❌Not a core point.
H	G,I	❌Not a core point.
I	G,H	❌Not a core point.

DBSCAN Example

Step 2: Start a Cluster from Core Points

B is a core point, so it forms a new cluster (Cluster 1).

B's directly reachable points (A, C, G) are added to the cluster.

Current Cluster 1:

{B, A, C, G}

DBSCAN Example

Step 3: Expand the Cluster

- We check if any of the new members (A, C, G) are also core points.
- C and A are not core points (not enough neighbors), so we don't expand from them.
- G has only 2 neighbors (H, I), which is less than 3, so it is NOT a core point either.
- Since there are no more core points, the expansion stops here.

DBSCAN Example

Step 4: Identify Remaining Points

- D, E, and F are not connected to B's cluster and form another group where all are neighbors of each other.
- None of them are core points (each has only 2 neighbors), so they don't expand.
- They remain unclustered or noise, depending on the dataset structure.

DBSCAN Example

Final Clusters:

Cluster	Points
Cluster 1	B, A, C, G
Noise	D, E, F, H, I (since they never reached a core point)